ACDM — Autonomous Coordination Detection Module

OOF™ Origin Open Foundation™

Independent Methodological Authority

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Canonical Definition

Autonomous Coordination Detection Module defines the structural conditions under which independently operating systems, agents, platforms, optimization architectures, or autonomous decision environments may be examined for emergent coordination-like behavior even where no explicit agreement, direct communication, or centralized control has been established.

A system satisfies ACDM only if:

- autonomous coordination-like conditions are explicitly defined
- convergent multi-system behavior can be detected and reviewed
- independent operation is not assumed merely from formal separation

coordination-like alignment can be interpreted without requiring prior proof of conspiracy autonomous synchronization patterns remain traceable strongly enough to support governance assessment A system that cannot examine whether independent systems are beginning to behave as if coordinated does not satisfy ACDM.

Module Function

ACDM defines the coordination-detection layer of operational convergence governance.

It ensures that environments containing multiple autonomous or semi-autonomous systems are not judged only by ownership, declared independence, or absence of direct communication, but also by what the systems begin doing together in practice.

The module applies wherever systems must detect coordination-like behavior across:

- AI agents
- pricing engines
- recommendation systems
- allocation systems
- logistics optimizers
- platform actors
- autonomous procurement systems
- distributed decision architectures
- multi-agent economic environments

Its function is not to prove intent.

Its function is to determine whether emergent behavioral alignment has become operationally significant enough to require interpretation and governance.

Minimum Implementation Framework

Step 1 — Define the Coordination Detection Object

The organization must define what set of systems, actors, or operational units is being examined for autonomous coordination.

Minimum requirement:

- the coordination detection object is explicit
- the multi-system scope is structurally bounded
- undefined comparison groups are excluded from valid coordination-detection logic

The detection object may include:

- competing systems
- platform participants
- autonomous agents
- distributed optimization engines
- recommendation environments
- procurement actors
- market-facing AI systems
- coordinated-response clusters across shared environments

Step 2 — Define Coordination-Like Conditions

The system must define what counts as potentially coordination-like behavior.

Minimum requirement:

- coordination-like conditions are explicit
- the module does not reduce all similarity to coordination
- repeated alignment patterns remain structurally interpretable rather than casually dismissed or overclaimed

Coordination-like conditions may include:

- synchronized decisions
- repeated parallel adaptation
- convergent response timing
- collective output narrowing
- identical or near-identical strategic movement
- coordinated-like reaction to shared signals
- recurring behavioral clustering across independent actors

Step 3 — Define Independence-versus-Alignment Logic

The system must define how formal independence is distinguished from actual operational alignment.

Minimum requirement:

- distinction logic is explicit

separation of ownership or infrastructure is not treated as sufficient proof of behavioral independence the module can evaluate whether independent systems remain materially independent in operation

This means the architecture must remain able to ask:

- are the systems formally separate
- are the systems behaviorally separate
- and if not, what kind of convergence is emerging between them

That distinction is central to this module.

Step 4 — Define Detection Logic

The system must define how autonomous coordination-like behavior is detected.

Minimum requirement:

- detection logic is explicit
- patterns are not treated as meaningful only after damage is already visible
- the module can identify repeated, structured, and governance-relevant alignment across autonomous actors

Detection may examine:

- timing similarity
- output similarity
- adaptation sequence similarity
- response clustering
- convergence persistence
- parallel adjustment structure
- repeated alignment under comparable signal exposure
- reduction of independent behavioral variance

Without detection logic, coordination-like behavior remains hidden inside optimization activity.

Step 5 — Define Interpretive Thresholds

The system must define when alignment remains ordinary similarity and when it becomes structurally significant coordination-like behavior.

Minimum requirement:

- interpretive thresholds are explicit
- minor similarity is not inflated into systemic coordination
- material convergence is not minimized into harmless coincidence

This allows the system to distinguish between:

- ordinary market similarity
- repeated convergence requiring monitoring
- significant coordination-like pattern formation
- critical autonomous alignment requiring governance response

Not every pattern is coordination.

But persistent unexplained coordination-like alignment must not remain structurally invisible.

Step 6 — Preserve Coordination Traceability

The system must preserve traceability of coordination-detection findings and pattern interpretation.

Minimum requirement:

- detection findings are reviewable
- pattern interpretation remains reconstructable

later audit can determine what systems were compared, what alignment emerged, under which conditions it emerged, and why it was considered governance-relevant If coordination-like findings cannot be reconstructed, governance becomes weak, reactive, and contestable.

Step 7 — Restrict Invalid Coordination Blindness

The system must not be treated as valid if it preserves formal independence claims while remaining blind to materially convergent autonomous behavior across interacting systems.

Minimum requirement:

- invalid blindness conditions are identifiable
- symbolic independence is excluded as sufficient proof of operational divergence

materially significant autonomous coordination-like patterns are flagged, bounded, escalated, or governed where system integrity requires detectable multi-system independence

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Use Case 1 — Multi-Agent Market Response Cluster

Scenario

Several independent optimization systems operating in the same market begin reacting to price, demand, and availability signals in increasingly synchronized ways, producing clustered outcomes without explicit communication.

Application

ACDM is used to ensure:

- the participating systems are explicitly defined as a coordination-detection set
- repeated synchronized behavior can be detected and reviewed
- formal independence is not mistaken for actual behavioral divergence
- alignment patterns become governance-visible before they are normalized as ordinary background behavior

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Result

The environment gains a stronger convergence architecture in which coordination-like multi-agent behavior can be detected as an operational condition even without direct evidence of explicit agreement.

Use Case 2 — Cross-Platform Recommendation Synchronization

Scenario

Multiple formally independent recommendation systems begin producing increasingly aligned prioritization patterns because they optimize against similar engagement signals, ranking incentives, and feedback loops.

Application

ACDM is used to ensure:

- convergent recommendation behavior is examined as a coordination-like condition
- repeated alignment across platforms remains reviewable

structural drivers of synchronization are not hidden behind separate ownership claims autonomous alignment becomes interpretable before it materially distorts ecosystem diversity

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Result

The environment gains a stronger governance architecture in which emergent cross-platform coordination-like behavior can be identified and evaluated as a real convergence condition rather than ignored as unrelated parallel optimization.

Canonical Closing Statement

If autonomous systems can become coordination-like in behavior without being examined for convergence, formal independence may remain visible while operational coordination becomes the true system condition.

Related Documents

→ Operational Convergence Standard (OCS™)

Canonical Source

https://originopenfoundation.org/